

Problem G. Ohana Cleans Up

Time Limit 2000 ms
Mem Limit 262144 kB

Ohana Matsumae is trying to clean a room, which is divided up into an n by n grid of squares. Each square is initially either clean or dirty. Ohana can sweep her broom over columns of the grid. Her broom is very strange: if she sweeps over a clean square, it will become dirty, and if she sweeps over a dirty square, it will become clean. She wants to sweep some columns of the room to maximize the number of rows that are completely clean. It is not allowed to sweep over the part of the column, Ohana can only sweep the whole column.

Return the maximum number of rows that she can make completely clean.

Input

The first line of input will be a single integer n ($1 \leq n \leq 100$).

The next n lines will describe the state of the room. The i -th line will contain a binary string with n characters denoting the state of the i -th row of the room. The j -th character on this line is '1' if the j -th square in the i -th row is clean, and '0' if it is dirty.

Output

The output should be a single line containing an integer equal to a maximum possible number of rows that are completely clean.

Examples

Input	Output
4 0101 1000 1111 0101	2

Input	Output
3 111 111 111	3

Note

In the first sample, Ohana can sweep the 1st and 3rd columns. This will make the 1st and 4th row be completely clean.

In the second sample, everything is already clean, so Ohana doesn't need to do anything.